OFDM Project

Wireless Receivers: Algorithms and Architectures

Adrien Maillet González

Yassin Alnuaimee

École polytechnique fédérale de Lausanne

20th December 2024

EPFL

Adrien Maillet González – Yassin Alnuaimee

O U T L I N E

Introduction

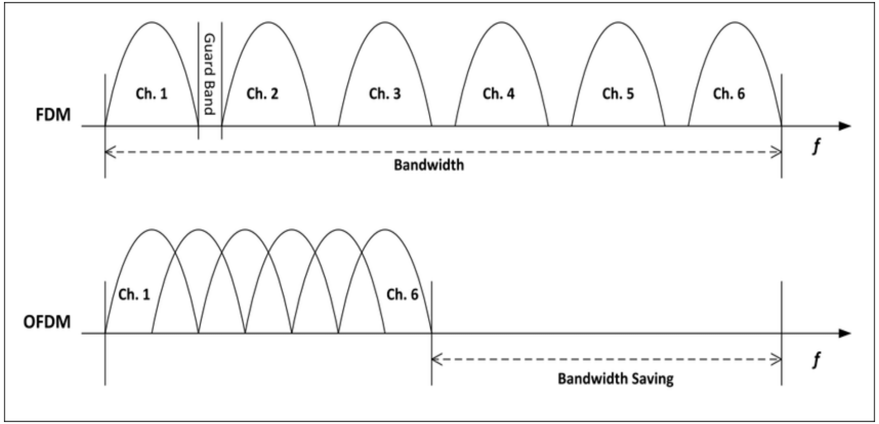
Implementation

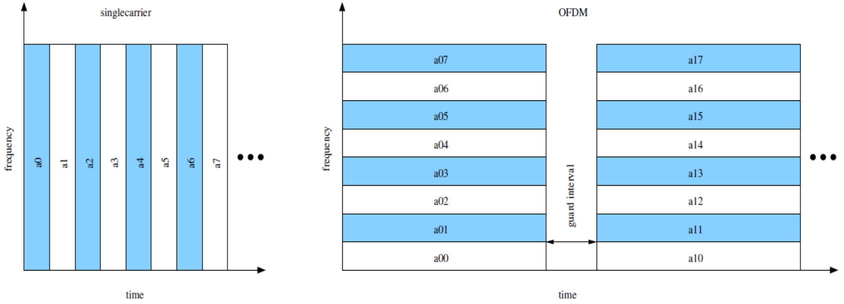
Results

Description of the project

- Implement an acoustic OFDM transmission system.
- **Goal:** send data through speakers and receive via microphone. Process and recover the data sent.
- **Subcarrier Division**: Bandwidth is divided into orthogonal subcarriers that overlap without interference, unlike FDM where guard bands are required in the frequency domain.
- **Symbol Transmission**: OFDM sends symbols in parallel, reducing the symbol rate and increasing symbol duration → mitigates ISI
- **Cyclic Prefix (CP)**: CP acts as a guard interval, transforming channel convolution into circular convolution, simplifying equalization.
- **Bandwidth Efficiency**: Overlapping orthogonal subcarriers save bandwidth, making OFDM more efficient than FDM.

→ **ISI Elimination**: Longer symbols + CP ensure that multipath effects do not cause interference between symbols.



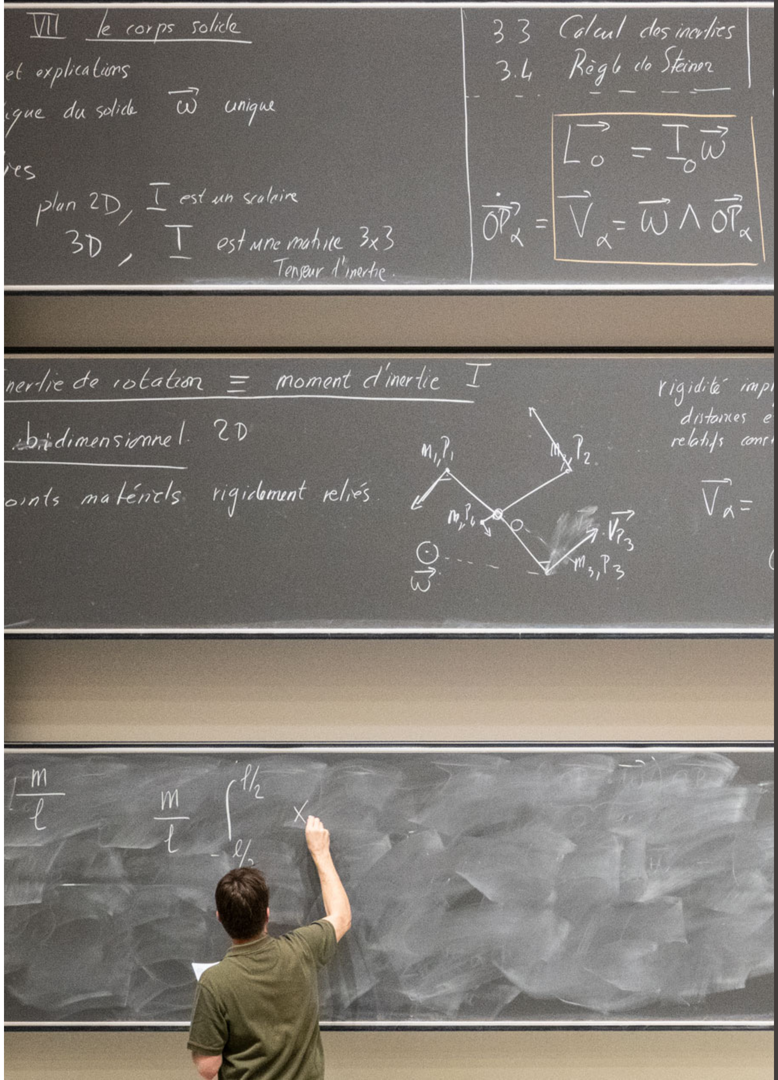


Adrien Maillet González – Yassin Alnuaimee

Why OFDM?

Adrien Maillet González – Yassin Alnuaimee

Method	Pros	Cons
Single Carrier Modulation	• Simple to implement • Low complexity for flat fading channels	• High ISI due to short symbol duration • Requires complex equalization in multipath • Sensitive to frequency-selective fading
Multicarrier Modulation	• Reduces ISI by splitting the bandwidth • Handles frequency-selective fading better	• Increased system complexity • Requires synchronization across subcarriers
OFDM (Orthogonal)	• Efficiently mitigated ISI using longer symbols • Simplifies equalization • Flexible for modern communication systems	• High Peak-to-Average Power Ratio (PAPR) • Sensitive to frequency offset errors • Requires adding cyclic prefix (overhead)



OUTLINE

Introduction

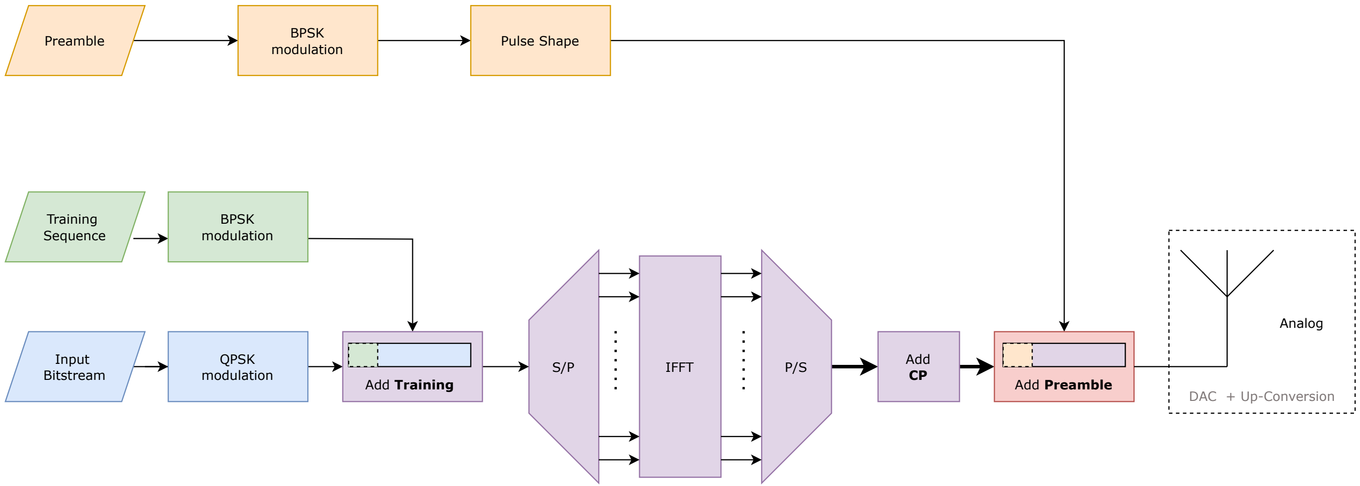
Implementation

Results

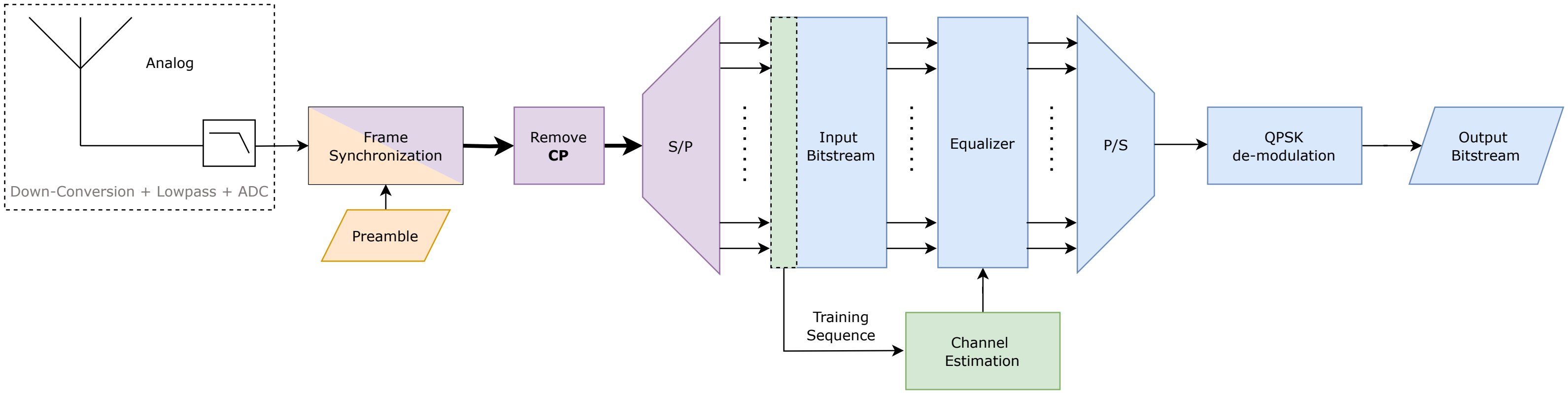
Adrien Maillet González – Yassin Alnuaimee

Adrien Maillet González – Yassin Alnuaimee

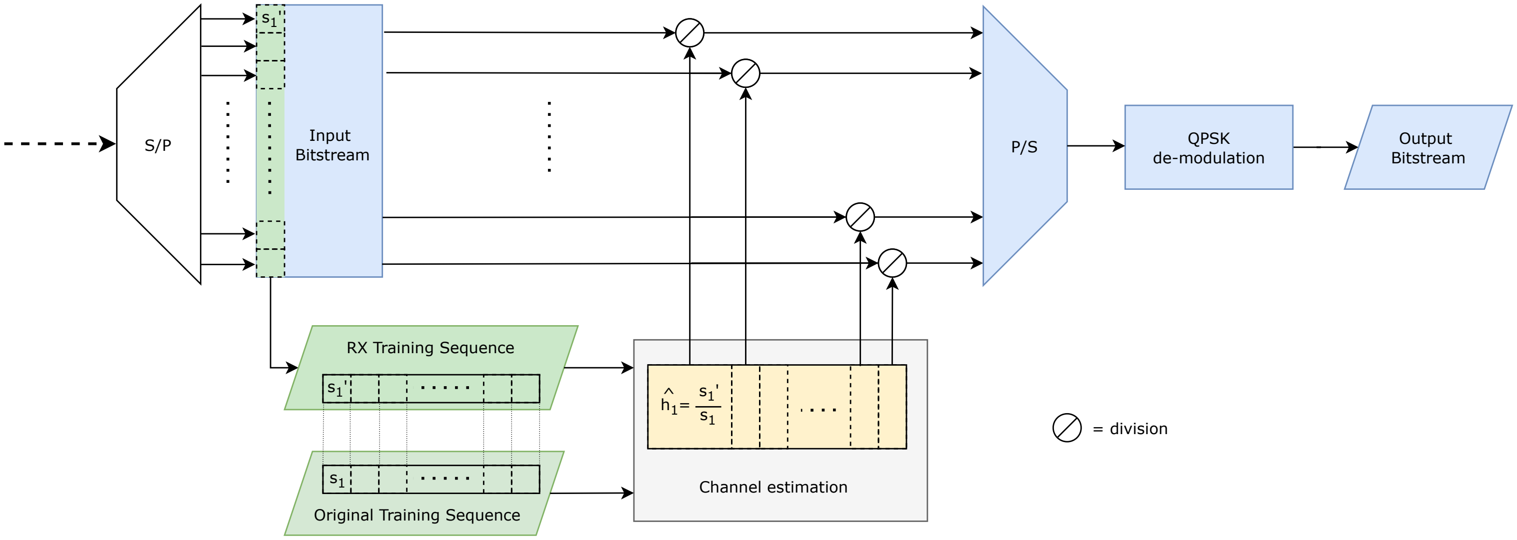
Transmitter



Receiver



Channel estimation



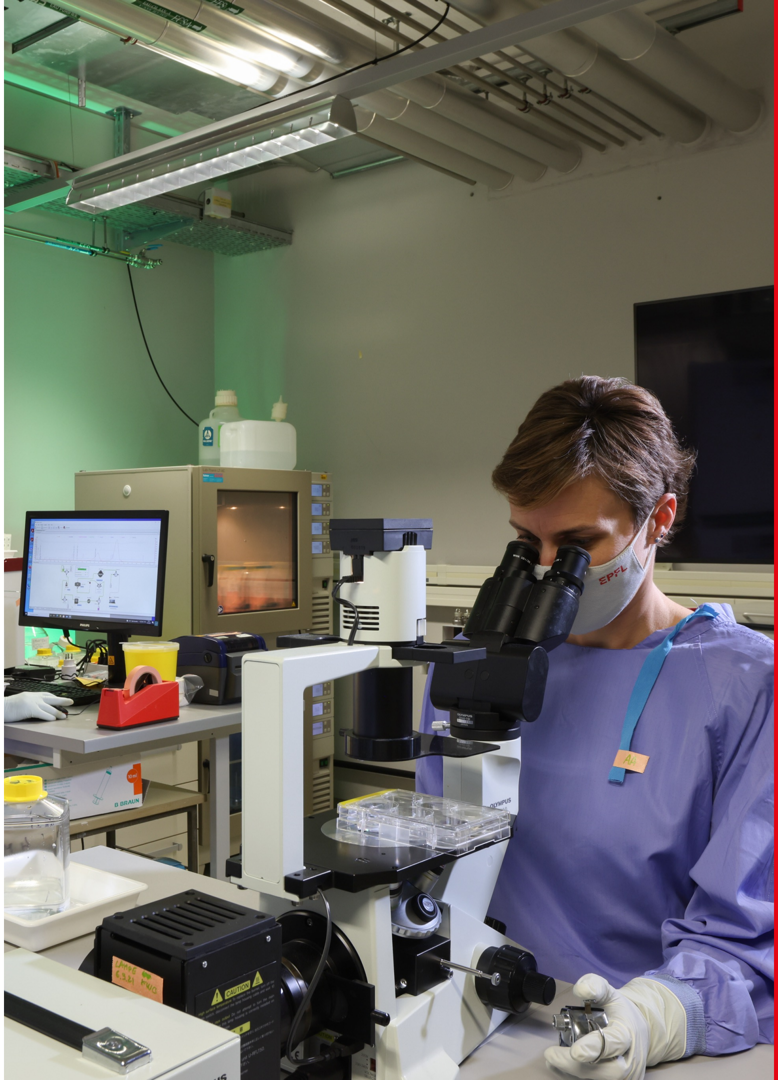
- We need the training the symbol to estimate the channel at the frequency of each sub-carrier

Phase tracking

Adrien Maillet González – Yassin Alnuaimee

	Continuous Phase tracking	Discrete Phase Tracking
Method	Viterbi and Viterbi Algorithm	Multiple Training Sequences (Block-Type Training)
Advantages	• No impact on the spectral efficiency • Works with no errors for a slow varying phase	• Can always recover from an abrupt phase shift
Disadvantages	• Not robust to abrupt phase shifts	• Decreases the efficiency • Unable to track the phase between two training sequences

Adrien Maillet González – Yassin Alnuaimee



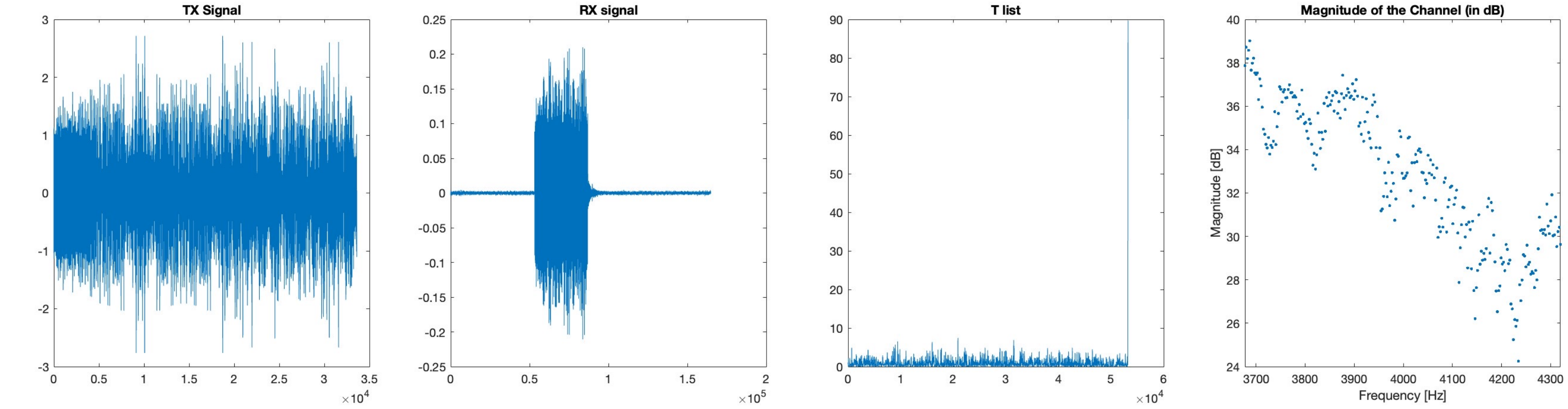
O U T L I N E

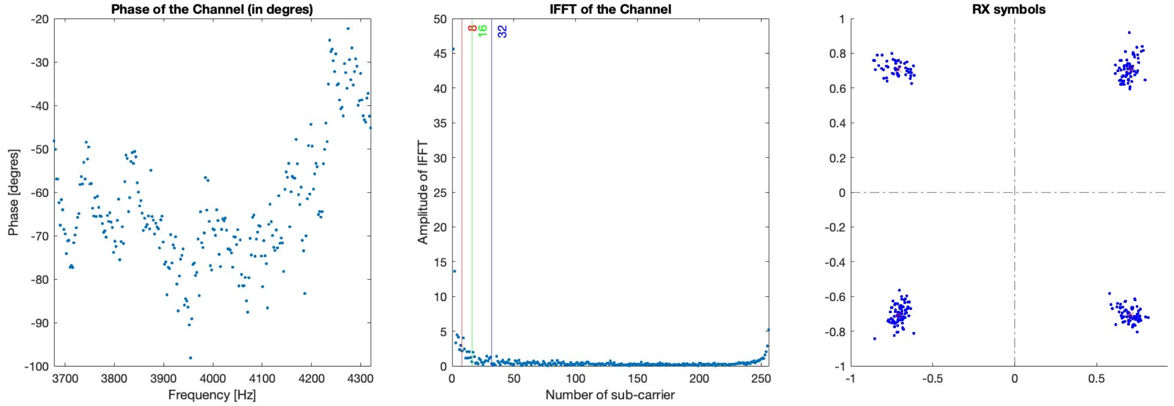
Introduction

Implementation

Results

Basic setup Results





$f_{carrier}$	4kHz
RX-TX distance	15 cm
# of sub-carriers	256
Δf (freq spacing)	5 Hz
length(CP)	256/2 = 128

Summary of the Basic Transmission

Adrien Maillet González – Yassin Alnuaimee

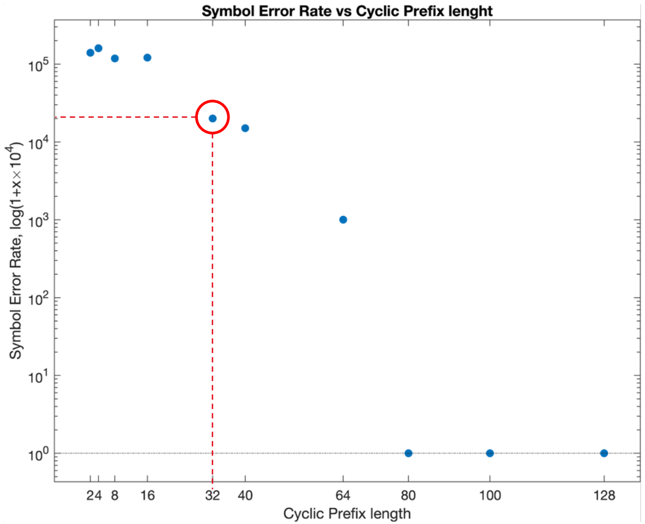
Cyclic Prefix

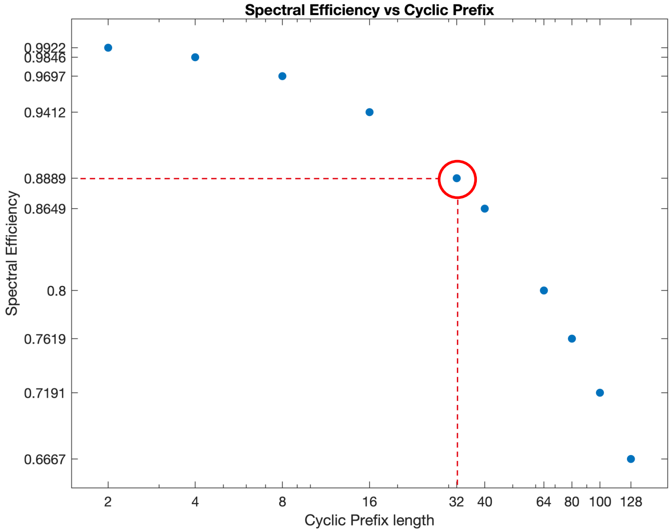
Number of sub-carriers	N = 256
Sub-carrier frequency spacing	Δf = 5Hz
Number of OFDM symbols sent	1

$lenght(CP_{original})$	$\frac{N}{2} = 128$
$length(CP_{required})$	32

$$\varepsilon = \frac{N}{N + lenght(CP)}$$

$len(CP_{required}) = 32 \rightarrow$ we get $\varepsilon = 89\%$

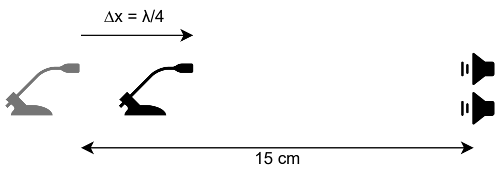




Adrien Maillet González – Yassin Alnuaimee

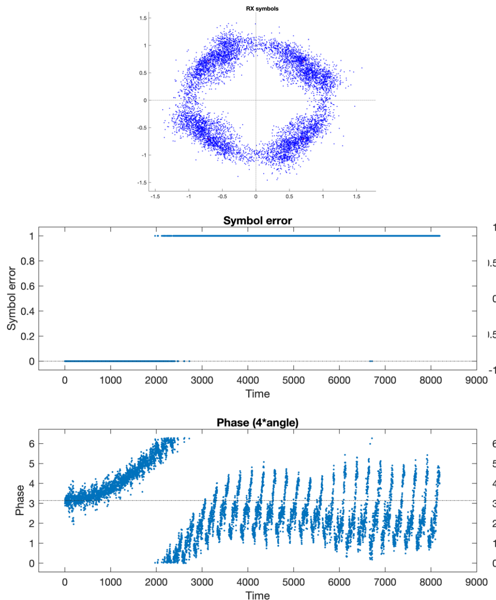
Phase Tracking: Moving System

- N = 256
- 32 symbols
- λ = 8.575 cm
- Δx = 2.1 cm

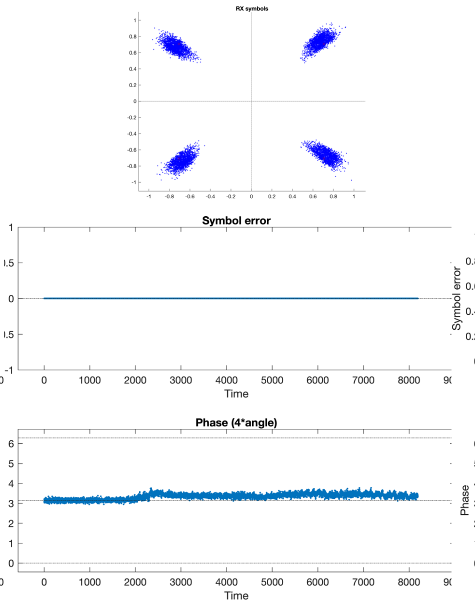


Adrien Maillet González – Yassin Alnuaimee

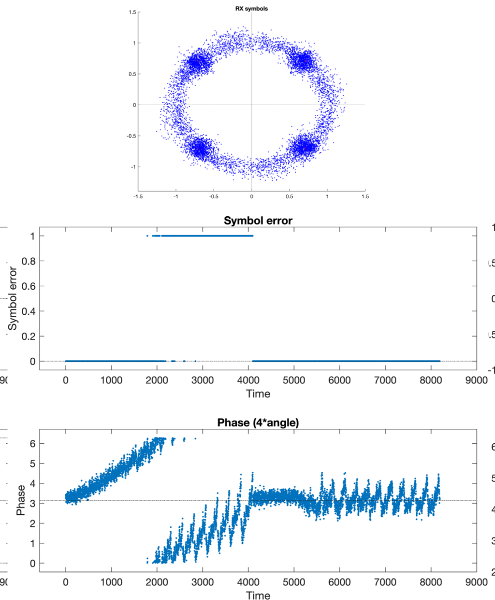
Without phase tracking



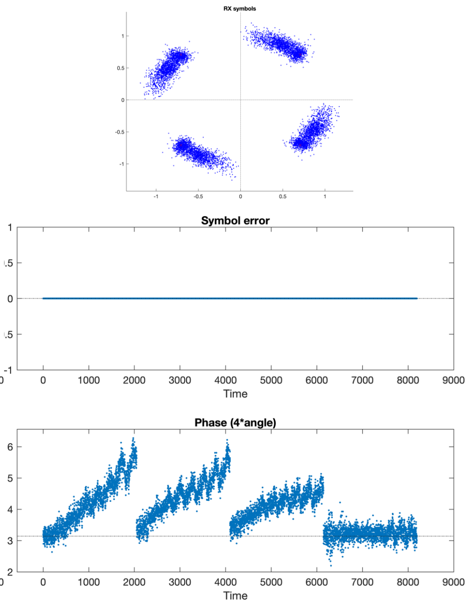
Continuous Phase Tracking



2 training symbols



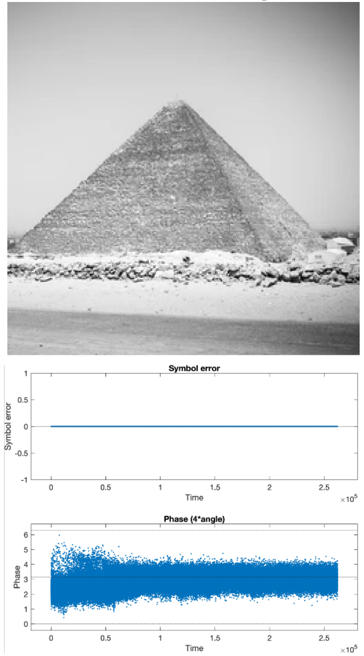
4 training symbols



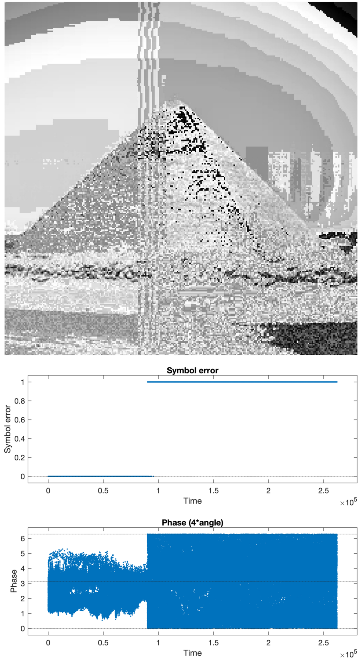
Real data transmission: Transmit Images

original image

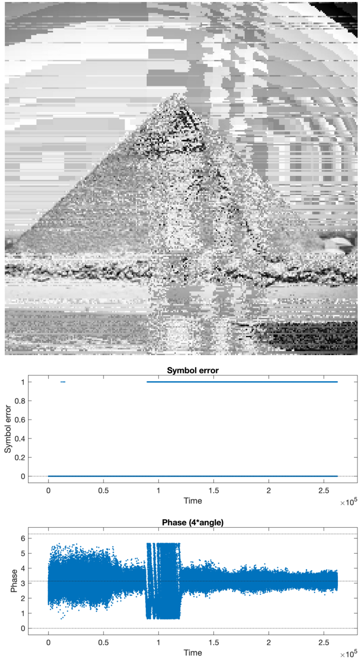
pyramid.png



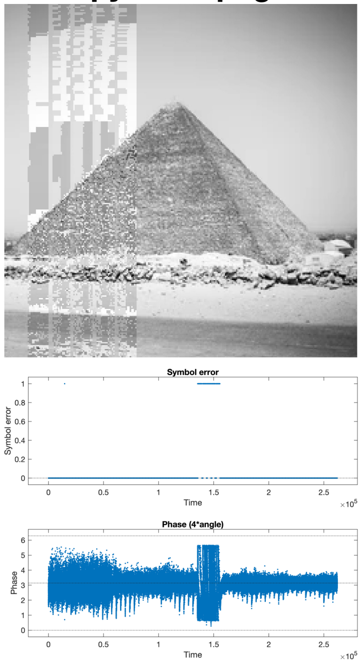
pyramid.png



pyramid.png



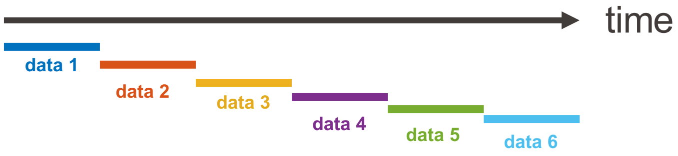
pyramid.png

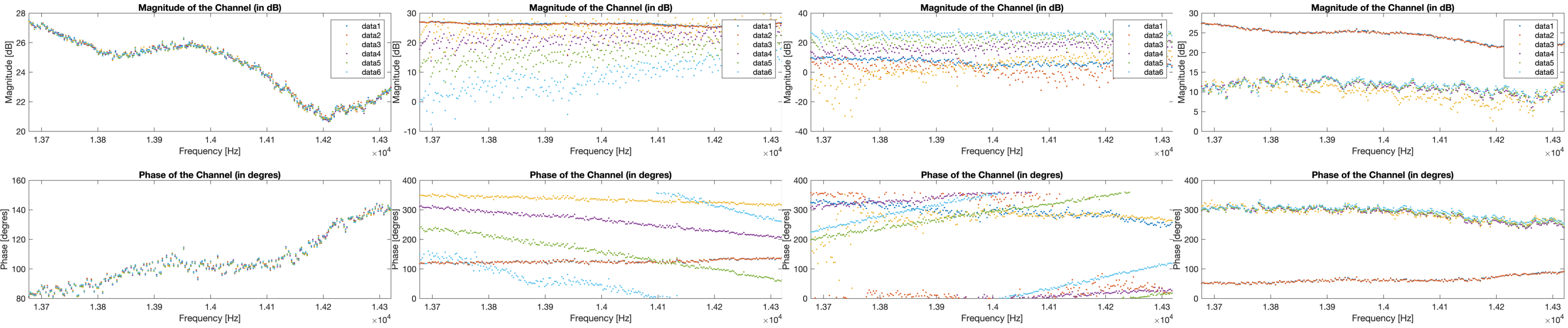


Number of subcarrier	256	256	256	256
Continuous Tracking	OFF	OFF	ON	ON
Multiple training	OFF	OFF	OFF	64 training sequences
Change setup distance	NO	1 λ of carrier (i.e., ~8.5cm)	1 λ of carrier (i.e., ~8.5cm)	1 λ of carrier (i.e., ~8.5cm)

Adrien Maillet González – Yassin Alnuaimee

Evolution of the Channel with time





Constant Channel | **Channel Size ↗️** | **Channel Size ↘️** | **Introduction of an obstacle ✋**

- As the change in the distance increases, we observe the slope of the Channel phases are becoming steeper
- As the distance increases the magnitude of the channel decreases
- As the distance decreases the magnitude of the channel increases
- As we introduce an object, we have a sudden drop in the magnitude of the channel, the phase however remains relatively constant along the change of frequencies

EPFL

Thank you

Adrien Maillet González

Yassin Alnuaimee

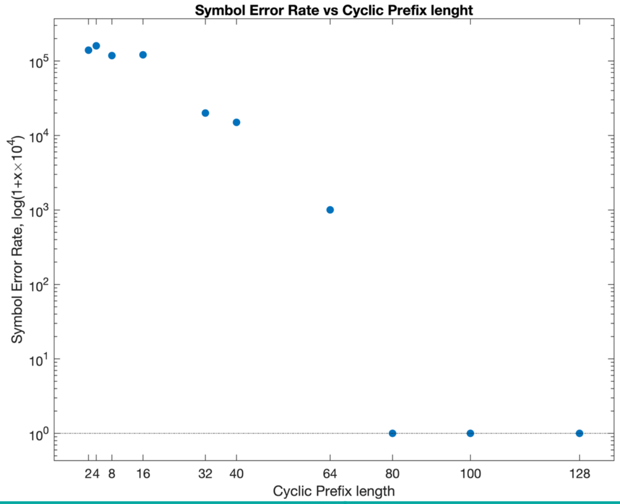
École
polytechnique
fédérale
de Lausanne

Adrien Maillet González – Yassin Alnuaimee

Backup

Cyclic Prefix

Number of sub-carriers	N = 256
Sub-carrier frequency spacing	Δf = 5Hz
Number of OFDM symbols sent	1



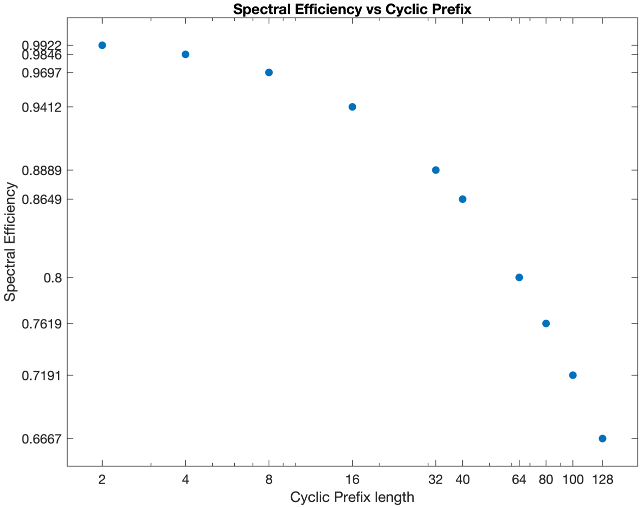
Adrien Maillet González – Yassin Alnuaimee

$$\varepsilon = \frac{N}{N + lenght(CP)}$$

for

$len(CP_{required}) = 32$

we get $\varepsilon = 89\%$



$lenght(CP_{original})$	$\frac{N}{2} = 128$
$length(CP_{required})$	32

Spectral efficiency: $\eta = \frac{R_b}{B}$

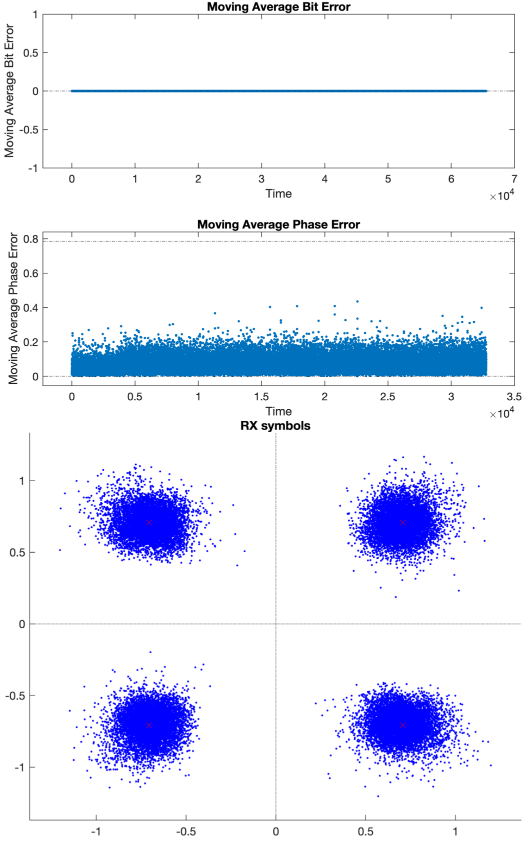
R_b: Bit rate
B: Bandwidth

$$\eta_{len(CP)=N/2} = \frac{\frac{256}{0.1+0.6}}{1280Hz} = 0.286[\text{symbol/s/Hz}] = \mathbf{0.571\ [bit/s/Hz]}$$

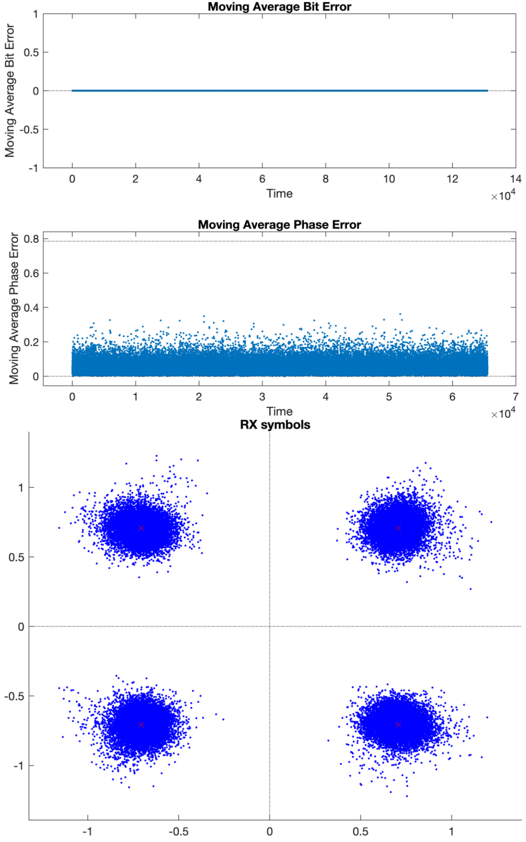
$$\eta_{len(CP')=32} = \frac{\frac{256}{0.1+0.45}}{1280Hz} = 0.364[\text{symbol/s/Hz}] = \mathbf{0.727\ [bit/s/Hz]}$$

NO Phase Tracking: Static System

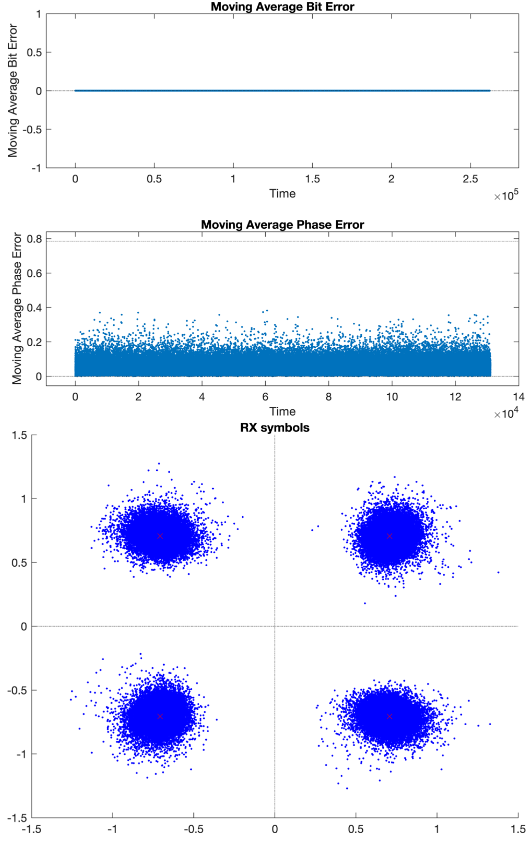
128 symbols



256 symbols

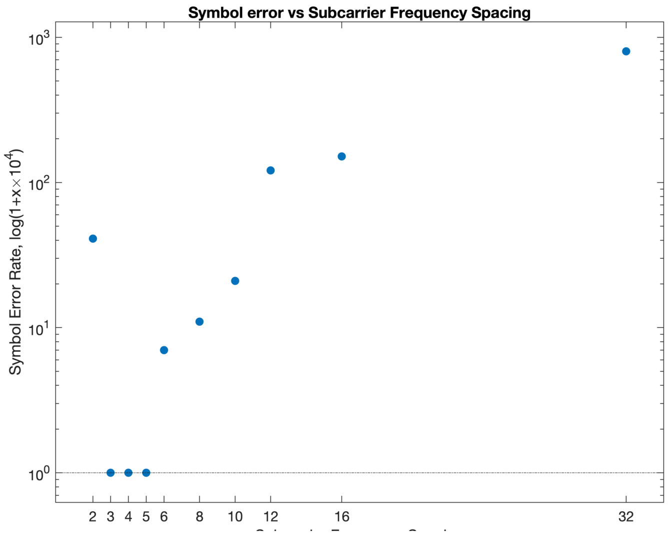


512 symbols

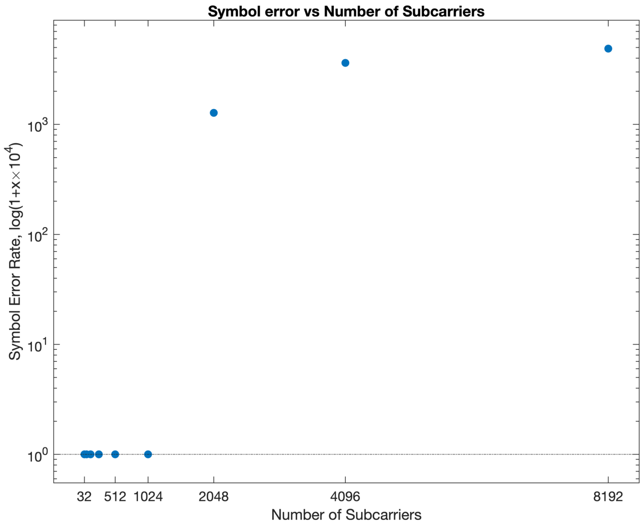


Adrien Maillet González – Yassin Alnuaimee

System performance evaluation



$f_{carrier}$	$4kHz$
Δx (distance microphone to speaker)	$2.00\ m$
N (number of sub carriers)	256
$length(CP)$	128
$\#OFDM\ symbols$	5



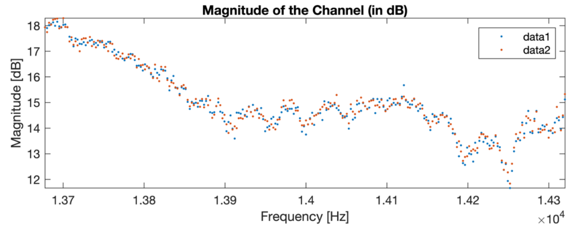
$f_{carrier}$	$4kHz$
Δx (distance microphone to speaker)	$2.00\ m$
Δf	5
$length(CP)$	$\frac{N}{2}$
$\#OFDM\ symbols$	5

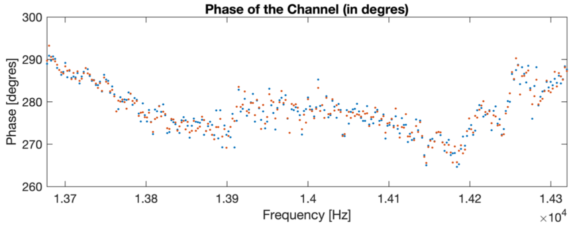
Adrien Maillet González – Yassin Alnuaimee

Channel Measurement

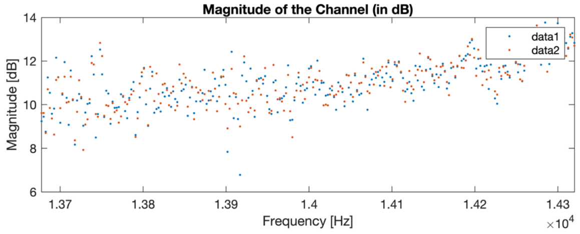
Best Channel:
- Short
- Calm
- Direct

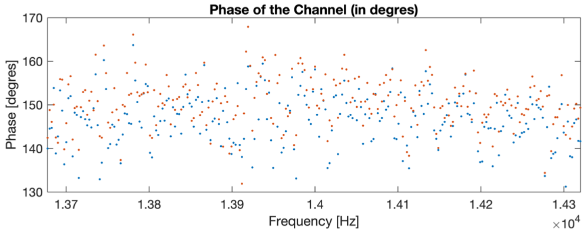
➢ High Gain
➢ Small Phase variations





Long Channel:
- Distance = 3m
- Calm
- Direct

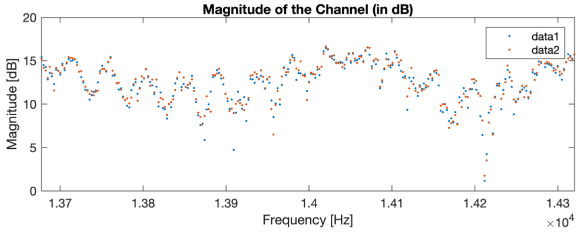


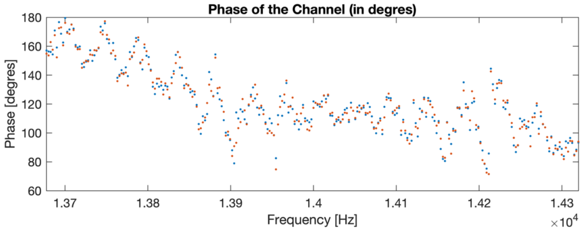


➢ -5dB magnitude
➢ Spread phase

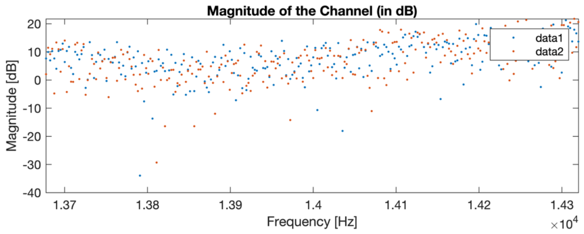
Reflection on Wall:
- Short
- Small incidence
- One path

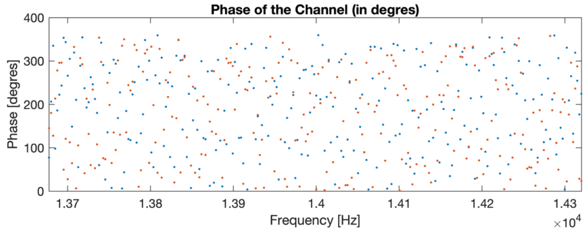
➢ High Gain
➢ Huge phase spread





Double Channel:
- Short
- Large incidence
- Two path to mic





➢ Lower gain
➢ Phases all around